

Optimal Climate Policy with Incomplete Markets

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What we do and why?

Introduction

- With no other distortions, the carbon tax equals the social cost of carbon: the **Pigouvian tax**
- With incomplete markets (inequality, risk) and distortionary taxes, the carbon tax interacts with other wedges

Should the carbon tax respond to inequality, risk, and fiscal constraints?

What we do and why?

Introduction

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Should the carbon tax respond to inequality, risk, and fiscal constraints?

- **Develop** a climate-economy model à la Barrage (2020) + the SIM model à la Aiyagari (1994)
- **Solve** the Ramsey problem:
 - carbon tax vs. Pigou
 - calibrate to US data
 - climate policy and welfare

What we find

Introduction

1. **Theory:**

- Distortions of intra and intertemporal margins **introduce a wedge** between Pigouvian rule and optimal carbon tax.

2. **Quantitative Model:**

- Optimal carbon tax **closely tracks social cost of carbon.**
- The Pigou principle **is robust and holds regardless of fiscal constraints.**
- True even when other instruments move allocations.

3. **Link to policy:**

- Relative to IPCC's high-emissions scenario the optimal policy leads to **4.7% welfare gains.**
- **Asymmetry** of welfare effects: postponing costs more than over-taxing.

Where we fit

Introduction

- Optimal climate policy with complete markets (e.g., Golosov, Hassler, Krusell, and Tsyvinski, 2014; Bovenberg and de Mooij, 1994; Barrage, 2020; Douenne et al, 2023)
 - **Our paper:** introduce incomplete markets
- Distributional effects of climate policy (e.g., Känzig, 2023; Fried et al, 2018 and 2024; Benmir and Roman, 2022)
 - **Our paper:** study optimal policy, analyze the transition, and account for welfare benefits of mitigation
- Optimal fiscal policy with incomplete markets (e.g., Conesa, Kitao, and Krueger, 2009; Dyrda and Pedroni, 2023)
 - **Our paper:** introduce climate change and study climate policy

The Model

Environment

Model

- Discrete time t ; population N_t , growth $n_{t+1} = N_{t+1}/N_t - 1$
- Sector TFP grows exogenously: $A_{i,t} = (1 + g_{A,t})A_{i,t-1}$ (economy on a balanced-growth path) [► Details](#)
- Ex ante heterogeneous households, type i : initial productivity $e_{i0} \in E$ and assets $a_{i0} \in A_0$; fraction α_i ($\sum_i \alpha_i = 1$)
- Idiosyncratic productivity $e_{it} \in E$ evolves stochastically; history e_i^t with probability $\pi_{it}(e_i^t)$
- Households save only in a risk-free asset, subject to a borrowing constraint
- Two production sectors: a final-good sector and an energy sector that generates CO₂ emissions
- Government levies distortionary taxes, issues debt, and sets climate policy

Preferences and budget

Model

Life-time expected utility of a type- i household:

$$\sum_t \beta^t N_t \sum_{e_i^t} \pi_{it}(e_i^t) [u(c_{it}(e_i^t), h_{it}(e_i^t)) - v(Z_t)]$$

- $\beta \in (0, 1)$: discount factor; c : consumption; h : labor
- $u(\cdot)$: per-period utility; $v(Z_t)$: utility damages from the climate Z_t (**increasing, convex**)

Per-capita assets evolve according to:

$$(1 + n_{t+1})a_{it+1}(e_i^t) = (1 + (1 - \tau_{K,t})(r_t - \delta))a_{it}(e_i^{t-1}) + T_t + w_t(1 - \tau_{H,t})e_{it}h_{it}(e_i^t) - c_{it}(e_i^t)$$

- w_t, r_t : before-tax wage and interest rate; $\tau_{K,t}, \tau_{H,t}$: capital / labor income tax; T_t : lump-sum transfer

Production: Final Goods

Model

- Technology:

$$Y_t = (1 - D(Z_t))A_{1,t}F(K_{1,t}, H_{1,t}, E_t)$$

→ Y : output

→ K : capital

→ H : labor

→ E : energy

→ $D(Z_t)$: production damages from environmental degradation Z (**increasing, convex**)

- Firm rents capital and hires labor and energy at factor prices r_t , w_t , and $p_{E,t}$, respectively

Production: Energy Sector

Model

- Technology:

$$E_t = A_{2,t}G(K_{2,t}, H_{2,t}),$$

- $G(\cdot)$: constant-returns-to-scale technology
- $K_{2,t}$ and $H_{2,t}$: capital and labor inputs fully mobile across sectors.
- $A_{2,t}$ denotes total factor productivity in the energy sector

- Energy production generates industrial CO₂ emissions,

$$E_t^M = (1 - \mu_t)E_t,$$

- μ_t : share of energy generated using clean technologies

- Profits:

$$\mathcal{P}_t = p_{E,t}E_t - \tau_{E,t}(1 - \mu_t)E_t - w_tH_{2,t} - r_tK_{2,t} - \Theta(\mu_t, E_t),$$

- $\Theta(\mu_t, E_t)$: abating emissions incurs a cost
- $\tau_{E,t}$: a tax on carbon emissions

Model

- Government budget:

$$G_t + N_t T_t + R_t B_t = \tau_{H,t} w_t H_t + \tau_{K,t} (r_t - \delta) K_t + \tau_{E,t} E_t^M + B_{t+1},$$

→ G_t : government spending; B_t : debt; R_t : gross interest rate on debt

- The climate variable Z_t :

$$Z_t = J(S_0, E_0^M, \dots, E_t^M, \eta_0, \dots, \eta_t).$$

- $\{E_t^M\}$: history of endogenous emissions
- $\{\eta_t\}$: other exogenous drivers
- S_0 : initial conditions

Theory

Ramsey problem

Theory

Utilitarian planner:

$$\mathcal{W}(\mathcal{T}) = \sum_i \alpha_i \sum_t \beta^t N_t \sum_{e_i^t} \pi_{it}(e_i^t) (u(c_{it}(e_i^t), h_{it}(e_i^t)) - v(Z_t))$$

Given K_0, B_0 , and the initial distribution over (a_{i0}, e_{i0}) , for each policy $\mathcal{T} = \{\tau_{Kt}, \tau_{Ht}, \tau_{Et}, T_t\}$:

- $X_i(\mathcal{T}) = \{a_{it+1}, c_{it}, h_{it}\}$: **individual** allocations (by type and history)
- $X(\mathcal{T}) = \{C_t, H_t, K_{t+1}, E_t, \mu_t, Z_t, B_{t+1}\}$: **aggregate** allocations
- $P(\mathcal{T}) = \{R_t, w_t, r_t, p_{E,t}\}$: **prices**

The Ramsey planner **announces and commits** to policies at date 0 and solves:

$$\max_{\mathcal{T}} \mathcal{W}(\mathcal{T}) \quad \text{s.t.} \quad \{X_i(\mathcal{T}), X(\mathcal{T}), P(\mathcal{T})\} \in CE(\mathcal{T})$$

- $CE(\mathcal{T})$: competitive equilibrium allocations and prices induced by policy $\mathcal{T}$

Pigouvian Tax ► Underlying Problem

Theory

The **Pigouvian tax** is defined as

$$\tau_{E,t}^{Pigou} = \frac{1}{\mathcal{W}_{c,t}} \sum_{j=0}^{\infty} \beta^j \left(\mathcal{W}_{c,t+j} D'_{t+j} A_{1,t+j} F_{t+j} - N_{t+j} \mathcal{W}_{Z,t+j} \right) J_{E_t^M, t+j},$$

where

- $\mathcal{W}_{c,t} = \sum_i \alpha_i \sum_{e_i^t} \pi_{it} u_{cit}$: the marginal utility of consumption, averaged across household types i and histories e_i^t
- $\mathcal{W}_{Z,t} = -v'(Z_t)$: the marginal utility damage from a higher temperature

Interpretation: the welfare cost of an extra CO₂ emitted at t , in units of date- t consumption, evaluated at a competitive-equilibrium allocation

Optimal Carbon Tax

Theory

Proposition

The Ramsey carbon tax satisfies a modified Pigouvian rule:

$$\tau_{E,t}^{SB} = \frac{1}{\nu_t} \sum_{j=0}^{\infty} \beta^j (\nu_{t+j} D'_{t+j} A_{1,t+j} F_{t+j} - N_{t+j} \mathcal{W}_{Z,t+j}) J_{E_t^M, t+j},$$

where ν_t is the multiplier on the resource constraint for the final consumption good satisfying:

$$\nu_t = \mathcal{W}_{c,t} + \sum_i \alpha_i \sum_{e_i^t} \pi_{it} (SD_{it} + LD_{it}),$$

with

$$SD_{i,t} \equiv u_{cc,it} \left(\kappa_{it}^s - \frac{R_t}{1+n_t} \kappa_{it-1}^s \right), \quad LD_{i,t} \equiv \kappa_{it}^\ell (u_{cc,it} w_t (1 - \tau_{H,t}) e_{it} + u_{ch,it}),$$

and where κ^s and κ^ℓ are the multipliers on the modified Euler equation and labor-supply FOC.

Optimal Carbon Tax Close by construction?

Theory

- Assuming $\{\tau_{H,t}, \tau_{K,t}, T_t\}_{t=0}^{\infty}$ are optimized:

$$\nu_t = \mathcal{W}_{c,t} + \underbrace{\sum_i \alpha_i \sum_{e_i^t} \pi_{it} (SD_{it} + LD_{it})}_{\text{impact of distortionary taxes}},$$

where

$$SD_{it} \equiv u_{cc,it} \left(\kappa_{it}^s - \frac{R_t}{1+n_t} \kappa_{it-1}^s \right) \quad (\text{Savings Distortions})$$

$$LD_{it} \equiv \kappa_{it}^\ell (u_{cc,it} w_t (1 - \tau_{H,t}) e_{it} + u_{ch,it}) \quad (\text{Labor Distortions})$$

→ $SD_{it} = 0$ if household is borrowing constrained in consecutive periods

→ $LD_{it} = 0$ if there are no income effects on labor supply

- Constraints on $\{\tau_{H,t}, \tau_{K,t}, T_t\}_{t=0}^{\infty}$ imply additional terms.

Taking the Model to the Data

Households

Calibration

- Preferences over consumption, labor, and climate:

$$u(c, h, Z) = \frac{(c^\gamma (1 - \varsigma h)^{1-\gamma})^{1-\sigma} + (1 + \alpha_z Z^2)^{\sigma-1}}{1 - \sigma}$$

- $\{\beta, \gamma, \sigma, \varsigma\}$ calibrated to match:
 - Capital-output ratio of 2.6 (NIPA, 2009–2019)
 - Average hours worked = 0.24 (CPS)
 - Intertemporal elasticity of substitution = 1/1.5
 - Average Frisch elasticity = 1.0
- α_z chosen so that 26% of damages directly affect utility (Barrage, 2019)

Calibration: Inequality and Income Risk

Quantitative Analysis

Cross-sectional distributions

	Bottom (%) 0–5	1st	2nd	Quintiles 3rd	4th	5th	Top (%) 95–100	Gini
Wealth								
Data	−0.5	−0.5	0.8	3.4	8.9	87.4	65.0	0.85
Model	−0.2	0.1	1.7	3.6	6.7	88.1	70.0	0.85
Earnings								
Data	−0.1	−0.1	3.5	10.8	20.6	65.2	35.3	0.65
Model	0.0	0.1	3.6	12.0	17.7	66.6	37.5	0.65
Hours								
Data	0.0	2.7	13.8	19.2	27.9	36.4	11.1	0.34
Model	0.0	0.4	11.4	26.1	28.3	33.9	8.9	0.35

Labor income, 1-year growth rate	Variance	Kelly skewness	Moors kurtosis
Data (Pruitt and Turner, 2020)	2.33	−0.12	2.65
Model	2.32	−0.13	2.65

Production

Quantitative Analysis

- Cobb-Douglas technology for both sectors

$$F(K_{1,t}, H_{1,t}, E_t) = K_{1,t}^\alpha H_{1,t}^{1-\alpha-\nu} E_t^\nu,$$

with $\alpha = 0.3$, and $\nu = 0.04$ (Golosov et al, 2014), and

$$G(K_{2,t}, H_{2,t}) = K_{2,t}^{\alpha_E} H_{2,t}^{1-\alpha_E},$$

with $\alpha_E = 0.6$ (Barrage, 2019)

- Abatement cost function adapted from [DICE 2023](#):

$$\Theta(\mu_t, E_t) = P_t^{\text{back}} \frac{\mu_t^{c_2}}{c_2} E_t,$$

→ P_t^{back} is the [backstop price](#): the marginal cost of abating the last unit of CO₂

► Backstop Price

Results

Policy experiments ► Computational Method

Results

We study a government that chooses a **time-varying carbon tax** under three fiscal regimes:

1. Fixed debt-to-GDP, fixed other taxes

→ Transfers adjust to balance the budget

2. Flexible debt-to-GDP, fixed other taxes

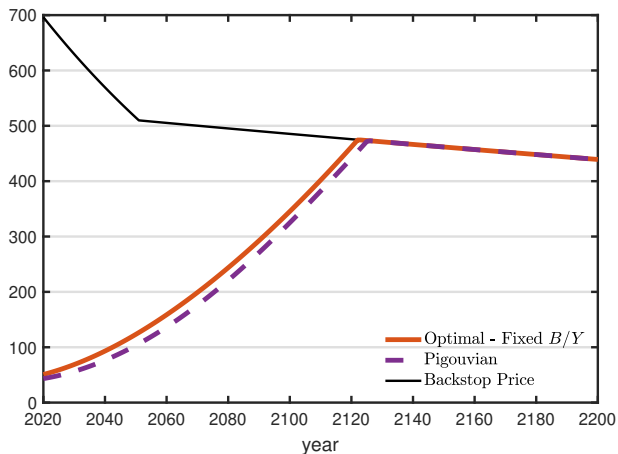
→ Transfers follow a flexible path: optimal to front-load transfers

3. Flexible debt-to-GDP, optimal constant labor and capital taxes

→ Optimal to increase both taxes $\tau_H = 27.7\% \rightarrow 41.5\%$, and $\tau_K = 33.6\% \rightarrow 44.7\%$

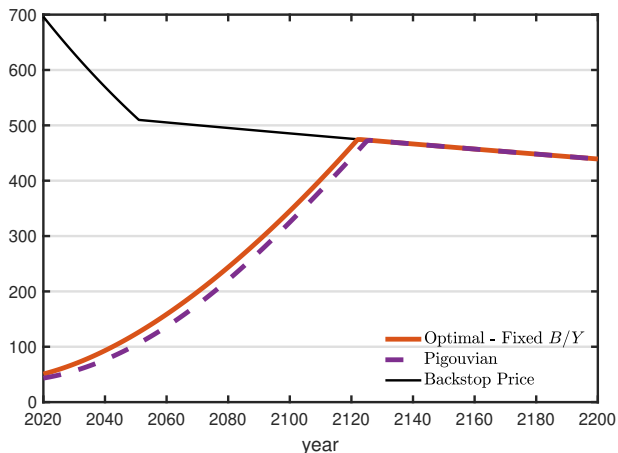
Optimal Carbon Taxes and Backstop Price (in $\$/\text{tCO}_2$)

Results



Optimal Carbon Taxes and Backstop Price (in $\$/\text{tCO}_2$)

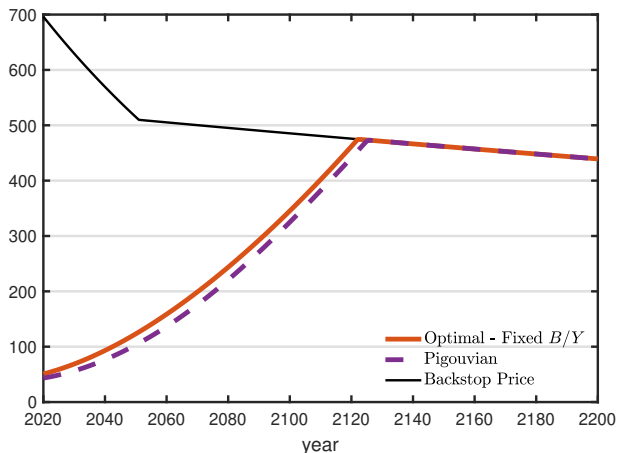
Results



Convex damages, cumulative temperature: each ton costs more than the last $\Rightarrow$ the Pigouvian path is convex.

Optimal Carbon Taxes and Backstop Price (in \$/tCO₂)

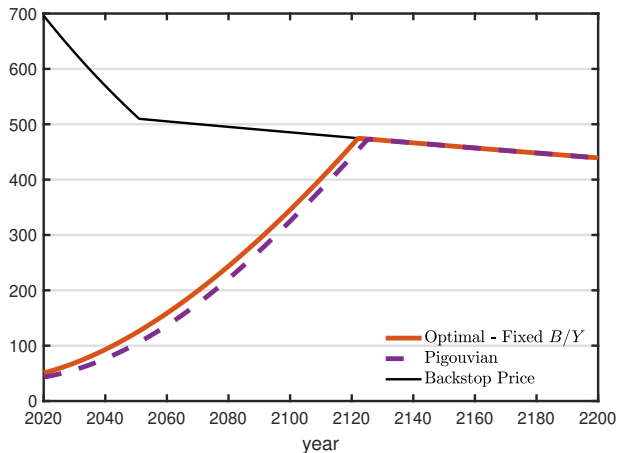
Results



At the own SCC, the climate margin is balanced; only fiscal and household responses can justify a wedge.

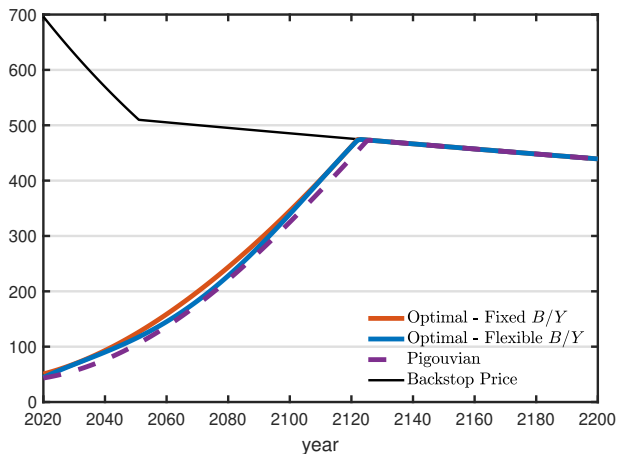
Optimal Carbon Taxes and Backstop Price (1/3)

Results



Optimal Carbon Taxes and Backstop Price (2/3)

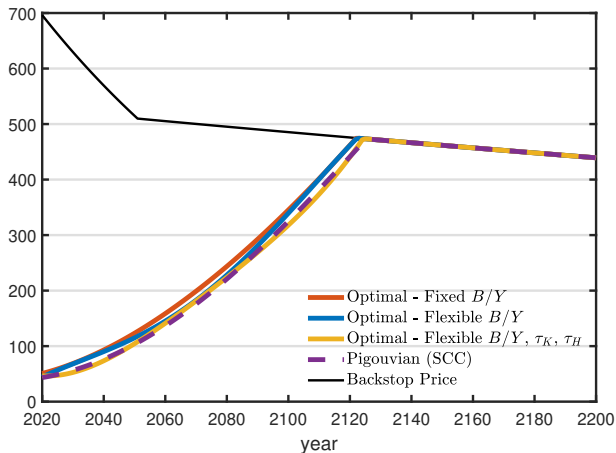
Results



Optimal Carbon Taxes and Backstop Price (3/3)

► Decomposition

Results



Debt, transfers, income taxes: however the fiscal side is relaxed, the tax tracks the SCC.

Why does the optimal tax track its own SCC?

Results

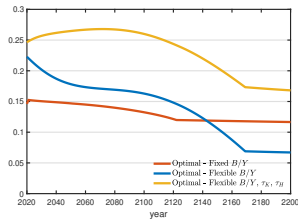
- Price the externality: set $\tau_{E,t}$ at the own SCC

$$\underbrace{\frac{\Theta_{\mu,t}}{E_t}}_{\text{marginal abatement cost}} = \tau_{E,t} = \underbrace{\frac{1}{W_{c,t}} \sum_{j=0}^{\infty} \beta^j (\mathcal{W}_{c,t+j} D'_{t+j} A_{1,t+j} F_{t+j} - N_{t+j} \mathcal{W}_{Z,t+j}) J_{E_t^M,t+j}}_{\text{discounted marginal damage}}$$

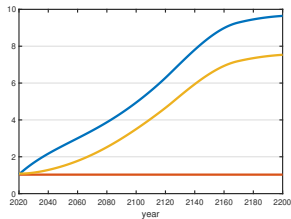
- A deviation moves the allocation (abatement, emissions $\rightarrow$ temperature) and creates a fiscal flow
 - $\rightarrow$ both reach households through $\{w_t(1 - \tau_{H,t}), (1 - \tau_{K,t})(r_t - \delta), T_t, Z_t\}$
 - $\rightarrow$ the Z_t leg is priced above $\Rightarrow$ residual first-order gains are fiscal
- **Free instruments** set after-tax prices and T_t directly $\Rightarrow$ welfare gains exhausted (envelope)
- **Fixed instruments**: the envelope fails along them
 - $\rightarrow$ a deviation mimics them through $(w_t, r_t, T_t) \Rightarrow$ value = multipliers on fixed instruments $\times$ quality of mimicry
 - $\rightarrow$ the elasticity of factor prices w.r.t. to the carbon tax governs the quality of mimicry (tied to energy's cost share) ► Magnitudes
- Welfare gain from optimal tax over Pigou: **0.008% (baseline), 0.001% (flexible instruments)**

Aggregates

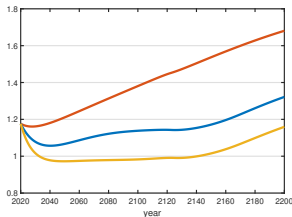
Results



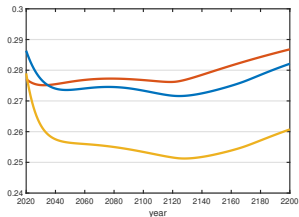
(a) Transfers to Output



(b) Debt to Output



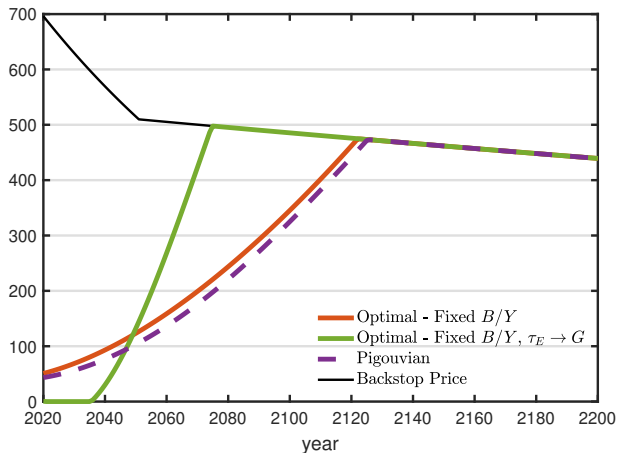
(c) Capital



(d) Consumption

Waste the revenue and the tax leaves the SCC

Results

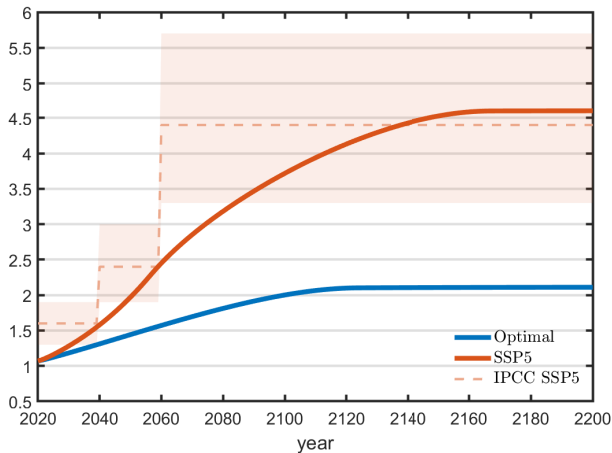


With revenue wasted, every dollar collected is a pure cost; revenue follows a carbon Laffer curve, peaking at intermediate rates $\Rightarrow$ the planner waits at zero, then jumps to the backstop.

Welfare Analysis

Shared Socioeconomic Pathway 5 (SSP5): Temperature

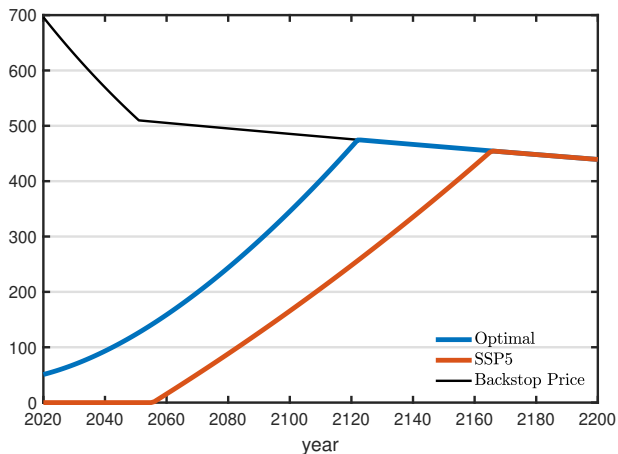
Results



Impose the SSP5 temperatures on the model by adjusting carbon taxes appropriately.

Corresponding Carbon Taxes (in $\$/\text{tCO}_2$)

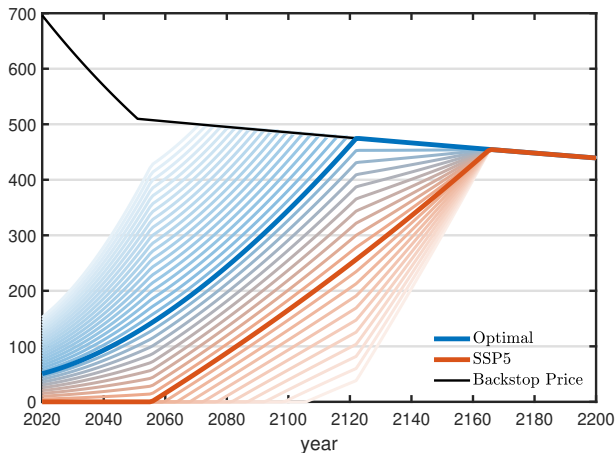
Results



Carbon taxes under SSP5 are postponed by two decades.

Varying Taxes From One to the Other

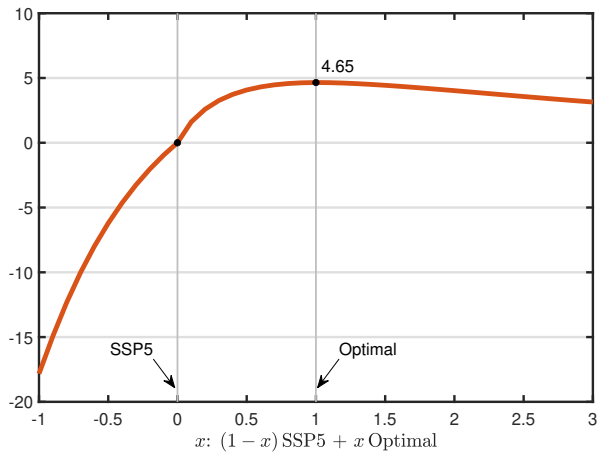
Results



Consider variation in the path of carbon tax according to: $(1 - x) \times \text{SSP5} + x \times \text{Optimal}$

Welfare Gain

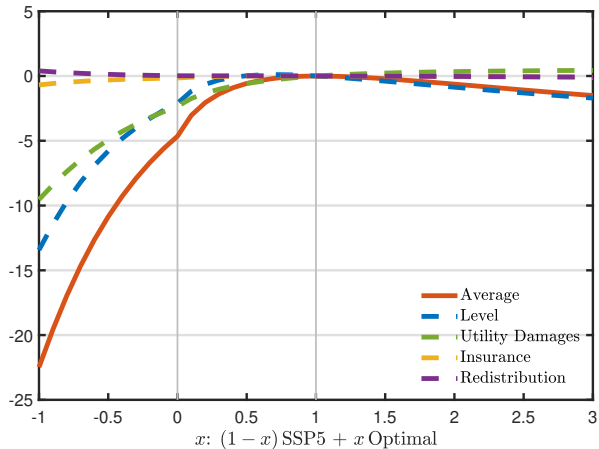
Results



Welfare falls faster below the optimum than above it.

Welfare Decomposition

Results



Welfare effects driven by efficiency (level) and utility damages; insurance and redistribution flat.

Key takeaways

Conclusion

Three insights:

1. **Theory**: the optimal carbon tax deviates from Pigou, with a wedge driven by savings and labor distortions
2. **Quantitatively**: the Pigou principle is alive and robust in the SIM-climate model, regardless of fiscal constraints
3. **Policy**: better to overshoot the carbon tax than to undershoot; welfare is steep to the left of the optimum, flat to the right

Extensions:

- Stone-Geary extension (in the paper): energy as a necessity; results hold. ► Stone-Geary
- In progress: non-linear tax schedule, non-separable damages, sensitivity w.r.t. to transfers/GDP.

Appendix

Theory & Mechanisms

Pigouvian Tax: Underlying Problem [◀ Back](#)

Appendix

Planner chooses allocations subject only to technology, resources, and the climate module:

$$\max_{\{c_{it}, h_{it}, K_{t+1}, E_t, \mu_t, Z_t\}} \sum_i \alpha_i \sum_t \beta^t N_t \sum_{e_i^t} \pi_{it} (u(c_{it}, h_{it}) - v(Z_t))$$

s.t. the resource constraint (multiplier $\beta^t \nu_t$) and the climate module (multipliers $\beta^t \lambda_{Z,t}$)

Three first-order conditions:

- Consumption: $u_{c,it} = \nu_t$ for all i , $e_i^t \Rightarrow$ marginal utilities equalized: $\nu_t = \mathcal{W}_{c,t}$
- Temperature: $\lambda_{Z,t} = \nu_t D'_t A_{1,t} F_t - N_t \mathcal{W}_{Z,t}$
- Abatement: $\nu_t \theta'_t(\mu_t) = \sum_j \beta^j \lambda_{Z,t+j} J_{E_t^M, t+j}$, where $\Theta_t(\mu_t, E_t) = \theta_t(\mu_t) E_t$

Energy firms set $\theta'_t(\mu_t) = \tau_{E,t}$; combining the three conditions yields the Pigou formula

Ramsey problem: same conditions for Z_t and μ_t ; household FOCs as constraints change the consumption condition: $\nu_t = \mathcal{W}_{c,t} + \sum_i \alpha_i \sum_{e_i^t} \pi_{it} (SD_{it} + LD_{it})$

Why deviations from Pigou are ≈ 0

[< Back](#)[MCF >](#)

Appendix

Cost of deviating ($W_{\tau\tau}$):

- Damages convex in Z ; Z linear in cumulative emissions, permanent
- Welfare falls by up to 22% of consumption toward SSP5

Benefit of deviating ($f_x x_\tau$):

- Transfers optimized: $f_x \approx 0$ (T^* balances equity vs. efficiency; envelope)
- Transfers constrained:

$$\tau^{**} - \tau^{Pigou} \approx - \frac{f_x x_\tau}{W_{\tau\tau}} - \frac{W_{\tau T}}{W_{\tau\tau}} (\bar{T} - T^*)$$

$\bar{T} - T^*$: room to improve; $W_{\tau T} \approx 0$: carbon-tax incidence flat (gains 4.6 to 5.8% across asset $\times$ income grid) $\Rightarrow$ no redistribution or insurance from τ

Representative agent: taxes distort with no equity or insurance motive $\Rightarrow \bar{T} - T^*$ and $W_{\tau T}$ both $\neq 0 \Rightarrow$ tax below Pigou (Bovenberg and Goulder, 1996; Barrage, 2020)

Marginal cost of public funds [◀ Back](#)

Appendix

Corollary (link with the MCF). The carbon tax is the modified Pigouvian rule adjusted for the MCF, the ratio of the resource-constraint multiplier to the average marginal utility of consumption:

$$\text{MCF}_t \equiv \frac{\nu_t}{\mathcal{W}_{c,t}} = 1 + \frac{1}{\mathcal{W}_{c,t}} \sum_i \alpha_i \sum_{e_i^t} \pi_{it} (SD_{it} + LD_{it})$$

- Deviations from Pigou arise whenever $\text{MCF} \neq 1$ (Barrage, 2020; Douenne et al., 2023)
- Complete markets: MCF averages to one under balanced growth $\Rightarrow$ Pigou on average (Douenne et al., 2023)
- Aiyagari economy: Ricardian equivalence breaks; MCF can lie above or below one
- Sign of $SD_{it} + LD_{it}$ is ambiguous: households may save or work too much or too little $\Rightarrow$ carbon tax above or below Pigou

Closeness to Pigou: Not by Construction

[< Back](#)[MCF >](#)

Appendix

- Evaluating both taxes at the same allocation fixes the units; the gap is governed by

$$\text{MCF}_t = \frac{\nu_t}{\mathcal{W}_{c,t}} = 1 + \frac{1}{\mathcal{W}_{c,t}} \sum_i \alpha_i \sum_{e_i^t} \pi_{it} (SD_{it} + LD_{it})$$

- Nothing in the definition pins $SD_{it} + LD_{it}$: multipliers on household FOCs at the optimum
- The same convention delivers gaps elsewhere:
 - Representative agent, constrained transfers: tax below Pigou (Bovenberg and Goulder, 1996; Barrage, 2020)
 - Revenue thrown away ($\tau_E \rightarrow G$): optimal tax at zero until the 2030s, then at the backstop
- Allocation dependence of the SCC: 2 to 10% across evaluation points; figures hold the allocation fixed
- Welfare metric independent of the convention: imposing Pigou costs 0.008% (baseline) and 0.001% (flexible instruments)

Why similar taxes across scenarios? [< Back](#)

Appendix

Optimal tax tracks the SCC; decompose it:

$$\tau_t^{e,Pigou} = \frac{1}{\lambda_t^A} \sum_{j=0}^{\infty} \beta^j \left(\lambda_{t+j}^A \lambda_{t+j}^B - \lambda_{t+j}^C \right) J_{E_t^M, t+j}$$

- λ^C (utility damages): pinned by temperature, similar across scenarios
- Debt and taxes crowd out capital, output falls:
 - $\lambda^B = D' A_1 F$ falls (production damages proportional to output; 74% of damages)
 - consumption falls, so $\lambda^A = \widetilde{\mathcal{W}}_c$ rises
- Damages enter as $\lambda^A \lambda^B$: fewer dollars of damage, each valued more
- $\sigma = 1.69$ (near log): offset close, not exact; SCC higher early, lower later

Competitive Equilibrium [◀ Back](#)

Appendix

Given K_0, B_0 , a distribution of types i over (a_{i0}, e_{i0}) , probabilities $\pi_{it}(e_i^t)$, populations $\{N_t\}_t$, and a tax policy $\mathcal{T} \equiv \{\tau_{H,t}, \tau_{K,t}, \tau_{I,t}, \tau_{E,t}, T_t\}_{t=0}^\infty$, a **competitive equilibrium** consists of individual variables $X_i \equiv \{a_{it+1}(e_i^t), c_{it}(e_i^t), h_{it}(e_i^t)\}_{i,t,e_i^t}$, aggregates $X \equiv \{C_t, H_t, H_{1,t}, H_{2,t}, K_{t+1}, K_{1,t}, K_{2,t}, E_t, \mu_t, Z_t, B_{t+1}\}_t$, and prices $P \equiv \{R_t, w_t, r_t, p_{E,t}\}_t$ such that: (i) households and firms optimize given prices and policies; (ii) the government satisfies its intertemporal budget constraint with bounded debt; (iii) temperature evolves according to (8); and (iv) markets clear: labor $H_t = H_{1,t} + H_{2,t}$ and capital $K_t = K_{1,t} + K_{2,t}$ split across sectors, and

$$C_t + G_t + K_{t+1} + \Theta_t(\mu_t)E_t = (1 - D(Z_t))A_{1,t}F(K_{1,t}, H_{1,t}, E_t) + (1 - \delta)K_t,$$
$$K_t + B_t = N_t \sum_i \alpha_i \sum_{e_i^{t-1}} \pi_{it-1}(e_i^{t-1}) a_{it}(e_i^{t-1}).$$

The economy is on a **balanced-growth path** if all aggregates except Z_t and μ_t grow at a constant rate and the equilibrium conditions hold.

Optimal Carbon Tax sits on Pigou: what is going on? [Details](#)

Results

A question of targeting. Planner chooses carbon tax τ and transfer T (financed with distortionary taxes):

$$W(\tau, T) = f(x(\tau, T), z(\tau)),$$

- x : economic outcomes (efficiency, redistribution, insurance); z : temperature

Distance to Pigou:

$$\tau^* - \tau^{Pigou} \approx - \frac{f_x(\tau^{Pigou}, T^*) x_\tau(\tau^{Pigou}, T^*)}{W_{\tau\tau}(\tau^{Pigou}, T^*)}$$

- Deviating pays only if **both** hold: τ moves outcomes ($x_\tau \neq 0$) **and** outcomes matter at the margin ($f_x \neq 0$)
- T optimized $\Rightarrow f_x \approx 0$; T constrained $\Rightarrow f_x \neq 0$ possible, yet $f_x x_\tau \approx 0$ quantitatively

How weak is the mimicry? Back of the envelope [◀ Back](#)

Appendix

- **Factor prices to the energy price:** substituting E out of F (Cobb-Douglas, K_1, H_1 fixed)

$$\frac{d \ln w}{d \ln p_E} = \frac{d \ln r}{d \ln p_E} = -\frac{\nu}{1 - \nu} = -0.042$$

→ w and r move equiproportionally: the bundle has one direction

- **Energy price to the tax:** $p_E = mc_E + \tau_E(1 - \mu) + \Theta/E$ and $\Theta_\mu = \tau_E E$

$$\frac{d \ln p_E}{d \ln \tau_E} = \frac{\tau_E(1 - \mu)}{p_E} \equiv s_\tau \approx 0.26 \quad (2020); \quad s_\tau \rightarrow 0 \text{ at the backstop}$$

- **Product:** $\frac{d \ln w}{d \ln \tau_E} = -\frac{\nu}{1 - \nu} s_\tau \approx -0.011$: doubling the tax moves w, r by 1%
- **Against the target:** freed, τ_H goes 27.7 to 41.5%: a 21 log-point after-tax wage move
 - requires $\Delta \ln \tau_E \approx 20$; room to the backstop: 2.6
 - revenue ceiling $s_\tau \nu \approx 1\%$ of output; transfer expansion when freed: 7 to 11pp of GDP

One direction, a tenth of the required move, vanishing at the backstop.

Decomposing the SCC [◀ Back](#)

Appendix

Decompose the Pigouvian tax (i.e., the SCC) as follows:

$$\tau_t^{e,Pigou} = \frac{1}{\lambda_t^A} \sum_{j=0}^{\infty} \beta^j \left(\lambda_{t+j}^A \lambda_{t+j}^B - \lambda_{t+j}^C \right) J_{E_t^M, t+j}, \quad (1)$$

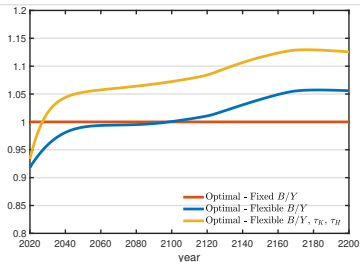
with

$$\lambda_t^A = \widetilde{\mathcal{W}}_{c,t} \quad ; \quad \lambda_t^B = D_t' A_{1,t} F_t \quad ; \quad \lambda_t^C = \mathcal{W}_{Z,t}. \quad (2)$$

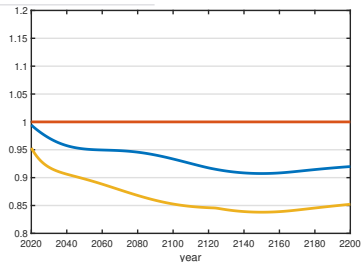
Components

[◀ Back](#)

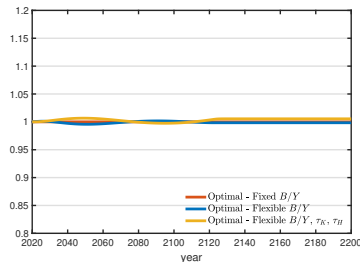
Appendix



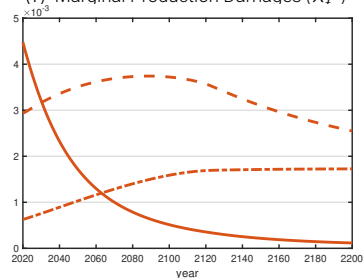
(e) Marginal Utility of Consumption



(f) Marginal Production Damages (λ_t^B)



(g) Marginal Utility from Climate (λ_t^C)



(h) Scale of Components

Welfare decomposition

Appendix

- The utilitarian welfare function can increase for four reasons:
 1. Reduction in distortions, if the utility of the average agent, $\sum_{t=0}^{\infty} \beta^t u(C_t, N_t)$, increases: **the level effect (Δ_L)**
 2. Lower utility damages from climate, $\sum_{t=0}^{\infty} \beta^t v(Z_t)$: **the utility-damages effect (Δ_{UD})**
 3. Transfers from ex-post rich to ex-post poor, if the risk of each individual path $\{c_t, n_t\}_{t=1}^{\infty}$ is reduced: **the insurance effect (Δ_I)**
 4. Transfers from ex-ante rich to ex-ante poor, if the inequality between certainty equivalents for $\{c_t, n_t\}_{t=1}^{\infty}$ is reduced: **the redistribution effect (Δ_R)**

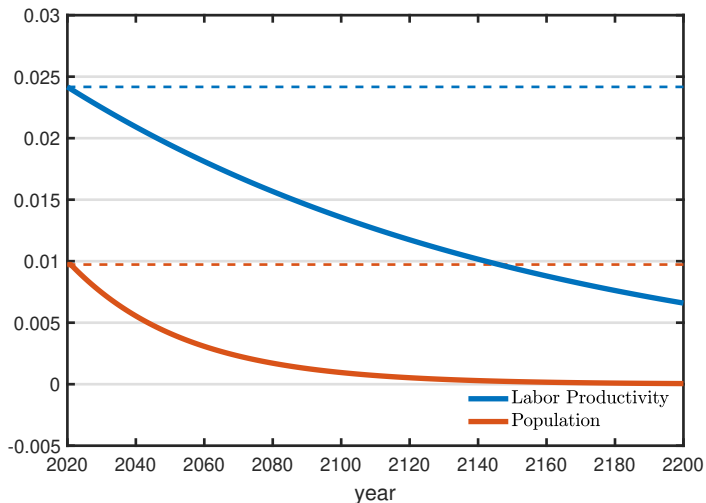
Let Δ be the utilitarian (average) welfare gain, which decomposes as:

$$(1 + \Delta) = (1 + \Delta_L) (1 + \Delta_{UD}) (1 + \Delta_I) (1 + \Delta_R)$$

Additional Results

Productivity and population growth over transition [◀ Back](#)

Appendix



Distribution of Gains

Appendix

Asset Quintile	Q1	Q2	Q3	Q4	Q5
	4.70	5.07	4.81	5.25	5.83
	4.59	4.60	4.58	4.58	4.59
	4.59	4.59	4.57	4.57	4.58
	4.58	4.58	4.57	4.57	4.57
Q1	4.58	4.57	4.56	4.56	4.58
Labor Income Quintile					

All households gain; gains rise with wealth and income.

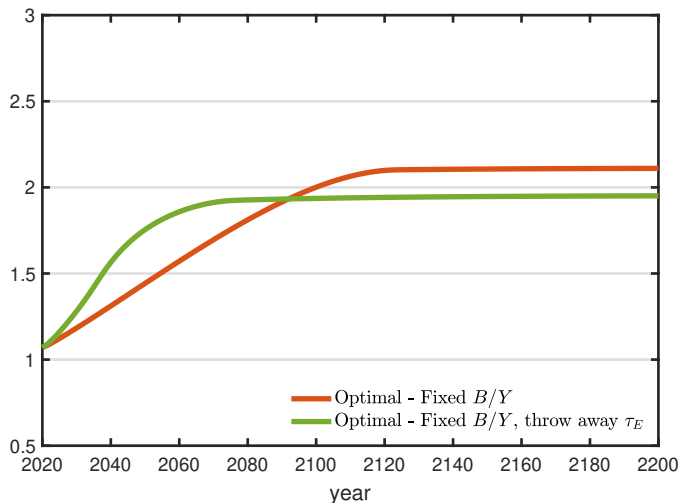
Backstop Price of Carbon Abatement [◀ Back](#)

Appendix

- The **backstop price** is the marginal cost of abating the last unit of CO₂.
- At this price, technology is available to:
 - Fully capture all emitted carbon, or
 - Produce carbon-free energy at scale to fully substitute fossil fuels.
- Key question: What is the **cost per ton of CO₂** for the cheapest such technology?
- The DICE model uses an **exogenous path** for the backstop price.
- This path is taken from a synthesis of external studies.
- It is **independent of the model's internal calibration**.

Optimal Carbon Taxes and Backstop Price (in $\$/\text{tCO}_2$)

Appendix



Backstop Price

Appendix

Barrage and Nordhaus 2024 (DICE 2023):

"The DICE model includes a backstop technology, which is a set of technologies that can replace all fossil fuels, albeit at a relatively high price. These technologies might be solar or wind power, safe nuclear power, or some as-yet-undiscovered source. Conceptually, at the cost of the backstop technology, the economy achieves zero net carbon emissions. Two revisions in the current version are noteworthy. Estimates of the cost of the backstop technology are controversial, with the DICE model having a high backstop cost relative to some estimates of the cost of renewables or carbon capture. The cost function is derived from highly detailed process models. Examining estimates of the marginal cost of scenarios with zero net emissions, we can estimate the marginal cost of the backstop technology. A statistical analysis from the results of the ENGAGE study (18, 19) indicates a median backstop price of \$15/ tCO_2 in 2019 in 2050, which is the earliest year that most models can reach zero net emissions. Models assume improvements over time in the technologies needed to attain zero emissions. The decline rate of the cost of the backstop technology is assumed to be 1%/year from 2020 to 2050, and then 0.1% year after that."

Calibration

Where we fit (2)

Appendix

Other papers on optimal carbon taxes in an Aiyagari-type models:

- Bourany (2024): cross-country heterogeneity; no other fiscal instruments
- Kubler (2024): constrained-optimal policy à la Dávila et al. (2012); theory-focused, abstracts from redistribution
- Belfiori, Carroll, Hur (2024): constrained-optimal policy; combines theory and quantitative analysis
- Malafry and Brinca (2022): early attempt at full welfare maximization; stylized (2 periods, no other instruments)
- Wöhrmüller (2024): richer model and careful calibration, but fixes other instruments and focuses on steady-state

Our paper: solve for dynamic optimal fiscal policy to maximize social welfare, with a rich calibration and more flexibility over the instruments

Climate Module: Dietz and Venmans (2019) and DICE 2023

Calibration

Temperature dynamics

- Global mean temperature:

$$Z_{t+1} = Z_t + \epsilon(\zeta \mathcal{E}_t - Z_t)$$

$$\zeta = 0.00045^\circ\text{C/GtCO}_2, \quad \epsilon = 0.5$$

- Cumulative emissions:

$$\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}$$

land emissions (DICE 2023) fall 10%/5yr

- Initial (IPCC 2021; GCP 2022): $Z_{2020}=1.1$, $\mathcal{E}=2390$,
 $E^M=36$, $E^{\text{ex}}=4$

Production damages

- Following Dietz and Venmans (2019):

$$D(Z_t) = 1 - \exp\left(-\frac{\alpha_1}{2} Z_t^2\right)$$

- Curve similar to DICE 2023
- $\alpha_1 = 0.01$: damages of 2% of GDP at 2°C , 7.7% at 4°C

Calibration: Macroeconomic Variables

Appendix

Macroeconomic aggregates		
	Target	Model
Intertemporal elasticity of substitution	0.66	0.66
Capital to output	2.57	2.54
Average Frisch elasticity (Ψ)	1.00	1.00
Average hours worked	0.24	0.25
Transfer to output (%)	14.7	14.7
Debt to output (%)	104.5	104.5
Fraction of hhs with negative net worth (%)	10.8	11.5
Correlation between earnings and wealth	0.51	0.43

Calibration: Population Partition

Appendix

		Shares			
		Population	Earnings	Income	Wealth
		Workers			
Data		67.2	82.7	69.1	44.9
Model		70.9	86.3	78.7	47.0
		Business Owners			
Data		5.8	13.7	16.1	33.0
Model		6.6	13.7	14.8	31.2
		Inactive Households			
Data		27.0	3.6	14.8	22.2
Model		22.5	0.0	6.5	21.8

Notes: Data comes from 2019 wave of the SCF

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Labor Productivity Process

Appendix

- Labor productivity has two components:
 - Persistent component e_P : Markov process with 4 states
 - Transitory component e_T : i.i.d. with 6 values
- Combined as: $e = e_P + e_T e_P^\eta$
 - $\eta = 0$: additive shocks; $\eta = 1$: multiplicative
- 26 free parameters after normalizing average productivity to 1
- Parameters calibrated to match inequality, risk, and population structure

► Population Partition

► Parameters

Labor Productivity Process: Parameters

Appendix

Parameter	Value
Curvature η	1.12

Persistent shock

$$\Gamma_P = \begin{bmatrix} 0.994 & 0.002 & 0.004 & 3\text{E}-5 \\ 0.019 & 0.979 & 0.001 & 9\text{E}-5 \\ 0.023 & 0.000 & 0.977 & 5\text{E}-5 \\ 0.000 & 0.000 & 0.012 & 0.987 \end{bmatrix} e_P = \begin{bmatrix} 0.185 \\ 0.305 \\ 0.537 \\ 27.223 \end{bmatrix}$$

Transitory shock

$$P_T = \begin{bmatrix} 0.357 \\ 0.002 \\ 0.467 \\ 0.004 \\ 0.025 \\ 0.176 \end{bmatrix} e_T = \begin{bmatrix} 0.07 \\ 0.09 \\ 3.12 \\ 3.16 \\ 7.80 \\ 9.51 \end{bmatrix}$$

Government

Appendix

- We extend procedure of Trabandt and Uhlig (2012) up to 2019:
 - labor income tax rate: $\tau_{H,0} = 27.7\%$
 - capital income tax rate: $\tau_{K,0} = 33.6\%$
- Debt (2019): difference between total liabilities and financial assets from the US government's balance sheet: $B_0/Y_0 = 104.5\%$
- Transfers capture all personal transfers from NIPA: $T_0/Y_0 = 14.7\%$

Exogenously Imposed Parameters (1)

Appendix

Parameter	Description	Value	Source
Production first sector			
a_1	Damage coefficient	0.0074	Dietz and Venmans, 2019 and Barrage, 2020
α	Return to scale on labor sector 1	0.3	DICE 2023
ν	Return to scale on energy sector 1	0.04	Golosov et al., 2014
δ	Depreciation rate on capital (per year)	0.1	DICE 2023
Y_{2020}	Initial output (in trillions 2023 USD)	83.476	World Bank (2016-2020)
Production second sector			
α_E	Return to scale on capital sector 2	0.597	Barrage, 2020
E_{2020}	Init. gross indus. emissions (GtCO ₂ per year)	38.23	Friedlingstein et al., 2022
Climate			
S_{2020}	Initial cumulative carbon emissions (in GtCO ₂)	2390	IPCC, 2021
T_{2020}	Initial atmos. temp. change (C since 1900)	1.07	IPCC, 2021
ϵ	Initial pulse-adjustment timescale	0.5	Dietz and Venmans, 2019
ζ	Trans. clim. resp. to cum. emissions (TCRE)	0.00045	IPCC, 2021
E_{2020}^{land}	Init. gross CO ₂ emis. land (GtCO ₂ per year)	4.17	Friedlingstein et al., 2022
ϵ_{land}	Ex. decline rate of gross land emissions (per period)	0.1	DICE 2023

Exogenously Imposed Parameters (2)

Appendix

Parameter	Description	Value	Source
Exogenous growth parameters			
$g_{A1,2020}$	Initial TFP growth rate sector 1 (per year)	0.0159	DICE 2023
$gg_{A1,t}$	Decline rate TFP growth sector 1 (per year)	0.0072	DICE 2023
$g_{A2,2020}$	Initial TFP growth rate sector 2 (per year)	0.0159	DICE 2023
$gg_{A2,t}$	Decline rate TFP growth sector 2 (per year)	0.0072	DICE 2023
N_{2020}	Initial population (in millions)	1,368	World Bank US-adjusted
$N_{\max}$	Asymptotic population (in millions)	1,910	DICE 2023 US-adjusted
g_N	Rate of convergence of population	0.145	DICE 2023
Fiscal Policy			
τ_K	Capital income tax (%)	33.6*	Appendix ??
τ_H	Labor income tax (%)	27.7*	Appendix ??
τ_C	Consumption tax (%)	4.2*	Appendix ??
$\tau_{I,t}$	Energy tax (%)	0.0	Appendix ??
$\tau_{E,t}$	Initial carbon emission tax (%)	0.6	Appendix ??

Exogenous growth and detrending

[◀ Back](#)

Appendix

Exogenous trends.

- Population: $N_{t+1} = (1 + n_{t+1})N_t$
- Sector TFP: $A_{i,t} = (1 + g_{A,t}) A_{i,t-1}$, $i \in \{1, 2\}$, with $g_{A,t}$ declining ($g_{A,2020} = 0.0159$, DICE 2023)

Labor-augmenting representation. Final good $Y_t = A_{1,t} K_{1,t}^\alpha H_{1,t}^{1-\alpha-\nu} E_t^\nu$ is Cobb-Douglas, so TFP maps to a labor-augmenting trend:

$$\Psi_t \equiv A_{1,t}^{1/(1-\alpha-\nu)}, \quad \Psi_{t+1} = (1 + g_{t+1})\Psi_t$$

Detrending. Stationary variables $\tilde{c}_t = c_t/\Psi_t$, $\tilde{a}_t = a_t/\Psi_t$, $\tilde{w}_t = w_t/\Psi_t$, $\tilde{T}_t = T_t/\Psi_t$:

- Borrowing limit grows with technology, $\underline{a}_{t+1} = \bar{a} \Psi_{t+1}$, so a balanced-growth path exists given policy
- Solve the stationary economy with time-invariant methods

[▶ Numerical Method](#)

Climate Module

Temperature

Appendix

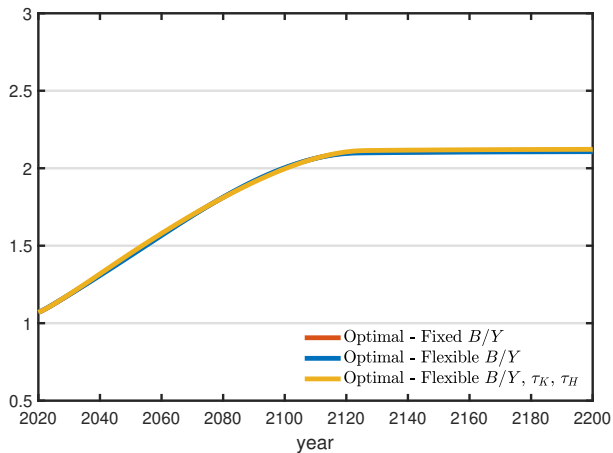
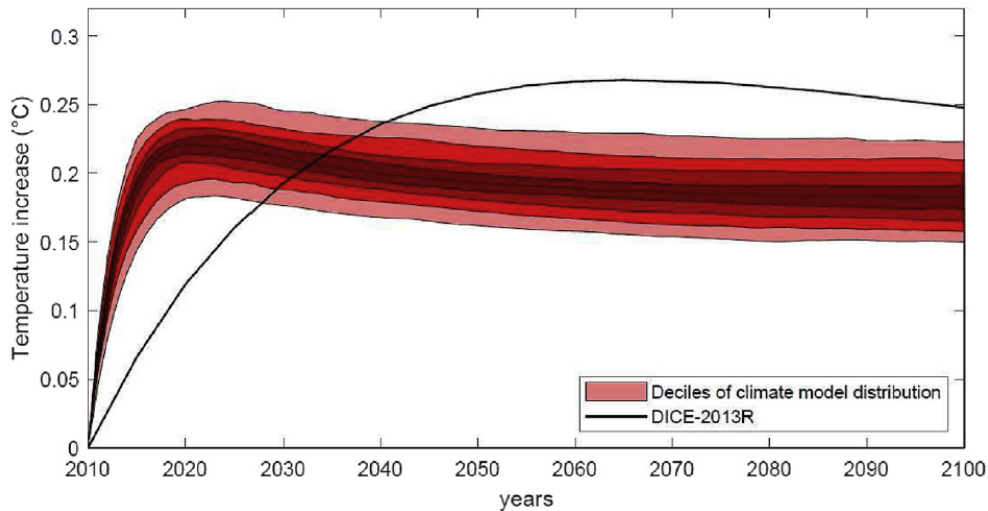


Figure: Temperature Change

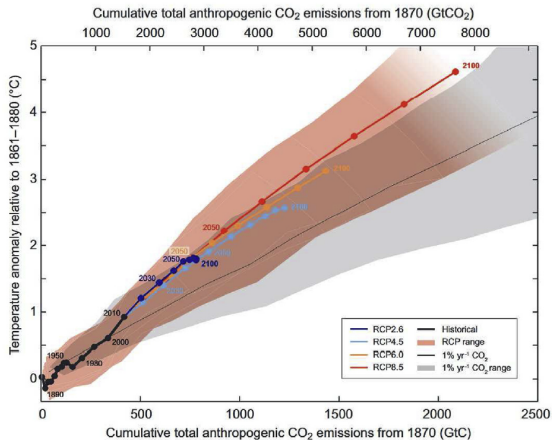
Temperature response to emissions

Appendix



Temperature linear in cumulative emissions

Appendix



Source: Dietz and Venmans (2019)

Extension: Stone-Geary

Extension: Energy Good and Stone-Geary Preferences ◀ Back

Appendix

- Poorer households emit less in total, but their consumption is more carbon-intensive (Sager, 2019): energy is a necessity
- Households consume the final good c and energy d directly:

$$u_t(c_{it}, d_{it}, h_{it}) = \frac{(c_{it} (d_{it} - \bar{d}_t)^\epsilon (1 - \varsigma h_{it})^\gamma)^{1-\sigma}}{1 - \sigma}$$

→ $\bar{d}_t = \bar{d} \Psi_t$: subsistence energy, grows with productivity (balanced growth)

→ Intratemporal condition: $u_d/u_c = p_{D,t} + \tau_{D,t}$, with $\tau_{D,t}$ a tax on household energy

- Energy market splits: $E_t = E_{1,t} + D_t$ (firms + households)
- Emission intensity ι of household energy: emissions $\propto E_{1,t} + \iota D_t$

Stone-Geary: Calibration [◀ Back](#)

Appendix

Choose $(\bar{d}, \epsilon)$ to match the distribution of household energy use (CEX, 2019):

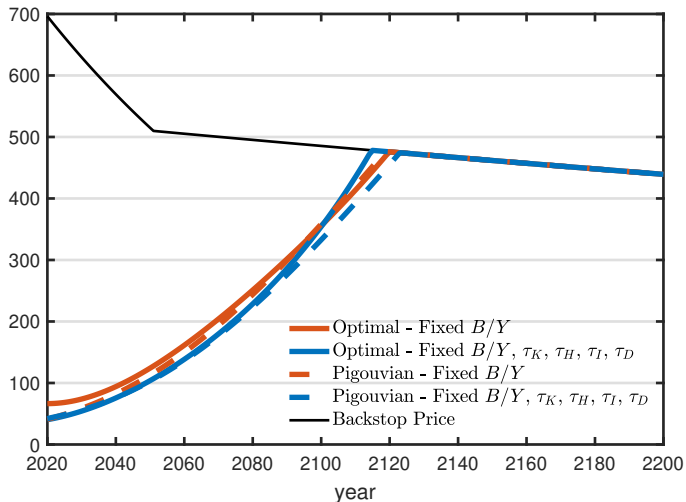
	Expenditure quintiles					Bottom/Top 5%	Gini
	1st	2nd	3rd	4th	5th		
Energy share (%)	11.3	16.6	20.0	23.3	28.9	1.8 / 8.5	0.178

- Household energy share of total expenditures: 7.0% (CEX, 2019)
- Household share of total emissions: 30% (EPA, 2024: residential + passenger vehicles)
- Remaining parameters: same procedure as the benchmark

Stone-Geary: Optimal Carbon Taxes (in $\$/\text{tCO}_2$)

[◀ Back](#)

Appendix



Optimal tax on the SCC in both scenarios; welfare cost of simply Pigou: 0.008% and 0.001%

Numerical Method

Computational method [◀ Back](#)

Appendix

- Detrend the level economy to a stationary system, then solve [▶ Detrending](#)
- We want to find the time paths $\{\tau_{K,t}, \tau_{H,t}, \tau_{E,t}, T_t\}_{t=0}^{\infty}$ that maximize welfare
- If optimal paths are smooth over time, we can approximate them with polynomials as in Dyrda and Pedroni (2023) [▶ Details](#)
- Polynomial parameters $\rightarrow$ path of fiscal instruments $\rightarrow$ transition to new balanced-growth path $\rightarrow$ welfare
- Optimize welfare by choosing polynomial parameters
- Bypasses the need to rewrite the Ramsey problem recursively

Computational Method: Details

Appendix

- Solving this problem involves searching on the space of sequences $\{\tau_{K,t}, \tau_{H,t}, \tau_{E,t}, T_t\}_{t=0}^{\infty}$
- To reduce the dimensionality of the problem, we follow Dyrda and Pedroni (2023) and parameterize the time paths of fiscal instruments:

$$x_t = \left(\sum_{j=0}^{m_{x0}} \alpha_j^x P_j(t) \right) \exp(-\lambda^x t) + (1 - \exp(-\lambda^x t)) \left(\sum_{j=0}^{m_{xF}} \beta_j^x P_j(t) \right)$$

- x_t can be any of the fiscal instruments $\{\tau_{H,t}, \tau_{K,t}, \tau_{E,t}, T_t\}$
- $\{P_j(t)\}_{j=0}^{m_{x0}}$ and $\{P_j(t)\}_{j=0}^{m_{xF}}$ are Chebyshev polynomials
- $\{\alpha_j^x\}_{j=0}^{m_{x0}}$ and $\{\beta_j^x\}_{j=0}^{m_{xF}}$ are weights on the consecutive elements
- λ^x controls the convergence rate of the fiscal instruments